Complete Concept Guide

BRFSS 2024 — Social Determinants of Mental Health

FIRST — What is Your Project About?

Say this to your professor:

> "My project analyzes what social and demographic factors predict poor mental health days using the CDC's BRFSS 2024 survey. I focused on 33 states that collected social determinant data — things like loneliness, emotional support, and life satisfaction — alongside traditional factors like sex, age, income, and smoking. My outcome variable is MENTHLTH — the number of days in the past 30 days when a person's mental health was not good."

Key numbers to remember:

- Dataset: **200,326 respondents** from 33 states
- Outcome variable: **MENTHLTH** (0 to 30 days)
- Mean mental health days: **~4.26 days**
- Median: **0 days** (most people report zero bad days)

CONCEPT 1: Why Did You Choose These 33 States?

What to say:

> "Not all 50 states collected the optional social determinant module — questions about loneliness (SDLONELY), emotional support (EMTSUPRT), and life satisfaction (LSATISFY) were only asked in 33 states. So I filtered the dataset to only include respondents from those states to ensure I had complete data for my key social variables."

Why this matters: This shows your professor you understood the survey design and made a thoughtful, deliberate data decision — not just used everything blindly.

CONCEPT 2: Descriptive Statistics

What they are: Basic summaries that describe your data before any testing.

Statistic	What it means	Your value
N	Number of observations	200,326
Mean	Average mental health days	~4.26 days
Median	Middle value	0 days
SD	How spread out values are	~8 days
Min / Max	Lowest and highest values	0 / 30

What to say:

> "The mean is 4.26 but the median is 0 this tells me the data is heavily right-skewed. Most Americans report zero poor mental health days, but a smaller group reports very high numbers, which pulls the mean up. This skewness is itself an important finding."

Key group differences to mention:

- Depressed respondents report **far more** poor mental health days than non-depressed
- "Always lonely" respondents report dramatically more days than "Never lonely"
- "Very dissatisfied" with life report more days than "Very satisfied"

CONCEPT 3: The t-Test

What it is: A test that compares the means of **exactly 2 groups** to see if they are significantly different.

Simple analogy:

> "Imagine weighing students from two different schools to see if one school has heavier students on average. A t-test tells you: is this difference real, or just due to random chance?"

You ran 2 t-tests:

t-Test 1: Sex vs Mental Health Days

- H0: Males and females have equal mental health days
- H1: They are different
- **Result:** Females report significantly more poor mental health days

- **What to say:** "Even after accounting for chance, females consistently report more poor mental health days than males. The p-value is essentially zero, meaning this result would almost never occur if there were truly no difference."

t-Test 2: Depression vs Mental Health Days

- H0: Depressed and non-depressed have equal mental health days
- H1: Depressed people have more
- **Result:** People with depression diagnosis report dramatically more poor mental health days
- **What to say:** "This result validates our outcome variable — if someone has been diagnosed with depression, they should logically report more poor mental health days. The fact that our test confirms this gives us confidence the data is measuring what it claims to measure."

CONCEPT 4: p-value and Hypothesis Testing

What it is: The p-value is the probability of seeing your results **if there were truly no difference** between groups.

The golden rule:

$p < 0.05$ → Result IS significant → Reject H0
$p > 0.05$ → Result is NOT significant → Keep H0

Simple analogy:

> "Imagine flipping a coin 100 times and getting 85 heads. The p-value asks: if this were a fair coin, how likely is it to get 85+ heads? If that probability is less than 5%, you conclude the coin is rigged. Same logic applies here."

In your project:

> "Every single test — both t-tests and all three ANOVAs — produced p-values essentially equal to zero. This means the probability of seeing these differences by chance alone is virtually zero. We confidently reject all null hypotheses."

Important caveat to mention:

> "With 200,000+ observations, even very small differences become statistically significant. So I also looked at the actual magnitude of differences — not just whether they were significant."

CONCEPT 5: ANOVA (Analysis of Variance)

What it is: ANOVA tests whether the means of **3 or more groups** are significantly different from each other — all at once.

Why not just use multiple t-tests?

> "If I tested all pairs of 5 loneliness groups individually, I'd need 10 separate t-tests. Running that many tests increases the chance of a false positive. ANOVA handles all groups simultaneously and controls for this."

How it works:

$$F\text{-statistic} = \frac{\text{Variance BETWEEN groups}}{\text{Variance WITHIN groups}}$$

- Large F = groups are very different from each other = significant

You ran 3 ANOVAs:

ANOVA 1: Loneliness vs Mental Health Days

- 5 groups: Always, Usually, Sometimes, Rarely, Never lonely
- **What to say:** "Loneliness showed the strongest ANOVA result. Those who are always lonely report dramatically more poor mental health days than those who never feel lonely. This confirms loneliness is a powerful social determinant of mental health."

ANOVA 2: Life Satisfaction vs Mental Health Days

- 4 groups: Very satisfied, Satisfied, Dissatisfied, Very dissatisfied
- **What to say:** "There is a clear gradient — as life satisfaction decreases, poor mental health days increase. Those very dissatisfied with life report the most poor mental health days."

ANOVA 3: Emotional Support vs Mental Health Days

- 5 groups: Always, Usually, Sometimes, Rarely, Never receive support
- **What to say:** "Emotional support acts as a protective factor. People who always receive emotional support report the fewest poor mental health days, while those who never receive it report the most. This highlights the value of social connections for mental wellbeing."

CONCEPT 6: Tukey HSD Post-hoc Test

What it is: After ANOVA tells you "at least one group is different," Tukey HSD tells you **exactly which pairs** are different.

Simple analogy:

> "ANOVA is like a fire alarm — it tells you there's a fire somewhere. Tukey HSD is the firefighter who finds exactly which room is burning."

What to say:

> "After my loneliness ANOVA showed a significant result, I ran Tukey HSD to find which specific loneliness groups differ. The test confirmed that every pair of loneliness levels is significantly different — meaning each step from Never to Always lonely corresponds to meaningfully more poor mental health days."

Why it's better than multiple t-tests:

> "Tukey HSD adjusts the p-values to account for making multiple comparisons, preventing false positives. This makes it more reliable than running individual t-tests for each pair."

CONCEPT 7: Confidence Intervals

What it is: A range of values we are 95% confident contains the true population mean.

Formula:

95% CI = Mean ± (1.96 × Standard Error)

Simple analogy:

> "A weather forecast says tomorrow will be 72-78 degrees. That range is a confidence interval — you're not sure of the exact temperature, but confident it falls in that range."

Key rule:

> "If the confidence intervals of two groups do NOT overlap, the groups are significantly different."

What to say:

> "My confidence intervals for depressed vs non-depressed respondents do not overlap at all — this visually confirms the difference is real and not due to chance. The narrow intervals also show that with 200,000+ respondents, our estimates are very precise."

CONCEPT 8: Spearman Correlation

What it is: Measures the **strength and direction** of the relationship between two variables. Values range from -1 to +1.

+1.0 = Perfect positive relationship
 0.0 = No relationship
-1.0 = Perfect negative relationship

Why Spearman instead of Pearson?

> "MENTHLTH is right-skewed — most values are zero with a long tail. Pearson correlation assumes normal distribution, which our data violates. Spearman works on ranks instead of raw values, making it appropriate for skewed and ordinal data."

Expected findings:

- **Loneliness** → strongest positive correlation with MENTHLTH (more lonely = more bad days)
- **Life Satisfaction** → strong positive correlation (less satisfied = more bad days)
- **Emotional Support** → negative correlation (more support = fewer bad days)
- **Age** → negative correlation (older = fewer bad days)
- **Income** → negative correlation (higher income = fewer bad days)

What to say:

> "The Spearman correlation heatmap revealed that social variables — loneliness, life satisfaction, and emotional support — have stronger correlations with mental health days than traditional demographic variables like age and income. This supports my hypothesis that social determinants are critical drivers of mental health."

CONCEPT 9: Multiple Linear Regression

What it is: A model that predicts an outcome (MENTHLTH) from **multiple predictors simultaneously**, while controlling for each other's effects.

Simple analogy:

> "Instead of asking 'does income affect mental health?' in isolation, regression asks: 'does income affect mental health even after accounting for age, sex, loneliness, depression, and everything else?' This gives us a much more accurate picture."

Your model:

MENTHLTH = Intercept
+ Sex coefficient
+ Age coefficient
+ Income coefficient
+ Smoking coefficient
+ Loneliness coefficient
+ Emotional Support coefficient
+ Life Satisfaction coefficient
+ Depression coefficient

Key results to mention:

Predictor	Effect	What it means
Depression	Largest positive	Being depressed adds the most bad days
Loneliness	Positive	More lonely = more bad days
Life Satisfaction	Positive	Less satisfied = more bad days
Emotional Support	Negative	More support = fewer bad days
Age	Negative	Older = fewer bad days
Income	Negative	Higher income = fewer bad days

What to say about R-squared:

> "My model's R-squared is higher than a demographics-only model because I included social determinants. Mental health is influenced by hundreds of factors, so explaining 10-15% of variance with 8 variables is meaningful. More importantly, every single predictor was statistically significant."

CONCEPT 10: Standard Deviation & Standard Error

Standard Deviation (SD)

What it is: How spread out individual values are around the mean.

What to say:

> "My MENTHLTH standard deviation of ~8 is larger than the mean of 4.26 — this confirms the data is highly spread out. Some people report 0 days while others report 30, creating very high variability. Lower income groups also show higher SD, meaning their mental health outcomes are more variable — some are fine but the worst cases are much worse."

Standard Error (SE)

What it is: How precisely your sample mean estimates the true population mean.

SE = SD ÷ square root of n

What to say:

> "Because my sample has 200,000+ observations, my standard errors are extremely small — meaning my mean estimates are very precise. This is why my confidence intervals are narrow and why even small differences are statistically significant."

THE BIG PICTURE — What Does Your Project Prove?

Opening statement for your professor:

> "My project tests a fundamental question in public health: are social factors — like loneliness, emotional support, and life satisfaction — stronger predictors of mental health than traditional demographic factors like age, income, and smoking?"

Key findings to deliver confidently:

1. **Depression is the strongest predictor** — as expected, those with a depression diagnosis report the most poor mental health days

2. **Loneliness is more important than income** — social isolation predicts mental health more strongly than financial status

3. **Emotional support is protective** — people who always receive emotional support have significantly better mental health outcomes

4. **Life satisfaction matters** — very dissatisfied people report dramatically more poor mental health days

5. **Young adults suffer the most** — ages 18-34 report more poor mental health days than any older group

6. **All results are statistically significant** — every test produced p ≈ 0, confirming these patterns are real and not due to chance

IF YOUR PROFESSOR ASKS TOUGH QUESTIONS

Q: Why is your R-squared low?

> "Mental health is influenced by hundreds of factors — genetics, relationships, trauma, sleep, and more. Explaining 10-15% of variance with 8 survey variables is actually meaningful in public health research. The focus is on which variables matter and in which direction, not on perfect prediction."

Q: Why did you use Spearman and not Pearson?

> "MENTHLTH is right-skewed with most values at zero. Pearson assumes normality, which this data violates. Spearman uses ranks instead of raw values and is the appropriate choice for skewed distributions."

Q: Can you prove causation?

> "No — this is observational, cross-sectional data. I can only establish associations, not causation. For example, loneliness may cause poor mental health, poor mental health may cause loneliness, or a third factor may cause both. This is an important limitation I acknowledge in my analysis."

Q: Why did you filter to 33 states?

> "The social determinant questions — loneliness, emotional support, life satisfaction are optional modules that only 33 states chose to administer in 2024. Including the other states would have resulted in 55%+ missing data for my key variables, which would severely compromise the analysis."

Q: Is your sample representative?

> "The BRFSS uses random-digit-dialing and has survey weights to adjust for sampling bias. The 33-state sample includes diverse regions and demographics, though it's important to note the findings strictly represent those 33 states rather than the full US population."

ONE-SENTENCE SUMMARY FOR YOUR PROFESSOR

> "This project demonstrates that social determinants — particularly loneliness, emotional support, and life satisfaction — are stronger predictors of poor mental health days than traditional demographic factors, using rigorous statistical methods including t-tests, ANOVA, Tukey HSD, Spearman correlation, and multiple linear regression on 200,326 respondents from the CDC's BRFSS 2024 survey."

You've got this! Every result in your project tells a clear, consistent story. Trust your data.